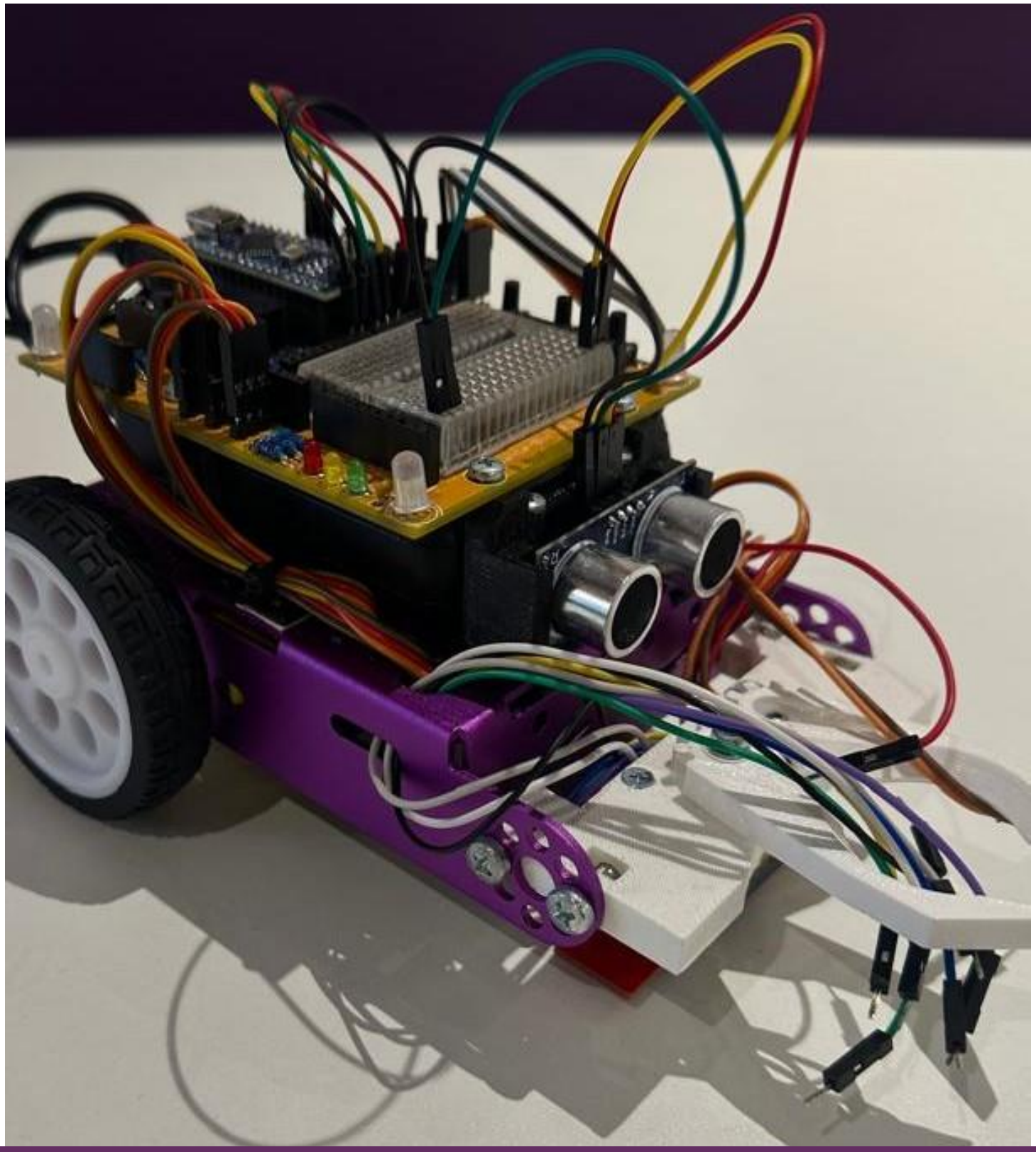
IT - 1B

PROJECT BATTLE BOTS



IVAN LYKHOTA, FRISO TERPSTRA, JEHAN MASTER, JOEL SHONIWA, KYRA KOVÁCS, MUHAMMAD UMAR KAMRAN

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1. Background:

Race Day is an event where students in the Information Technology course showcase their skills via the completion of a course with three bots working together to finish the course, while competing against other students based on how fast the course is completed

2. Goal and Result:

Program three different battle bots to complete a course with three separate parts. The separate parts are the physical maze, follow the line, and a maze but with black lines. Each bot must complete one of each as a relay race, while holding a cone, which will be used as a baton and placed in the black box at the end, which signifies that the course is completed.

3. Project Goal:

Program three individual bots to fully complete the three different courses, effectively while holding the cone, this can be done with motors and sensors, which should result in winning the robot race. This must be done in time for the for the Race Day in week 8.

4. Project Result:

The overall objective is to program three fully functional robotic vehicle which is made to navigate different environments in a relay-type time trial.

There are three different robots that have Arduino microcontrollers, which are the brains of the system. The system uses C/C++ programming to coordinate motors for propulsion and grippers with sensors that view the distance, a line sensor that detects the black tape, and sensors that involve the surrounding environment. Development is managed through Git version control, and code is being saved in personal portfolio websites and documented. Weekly assignments and final code should be seen inside said places. The final product must demonstrate the ability to follow lines, solve mazes, and interact with other robots to achieve the fastest and most accurate possible race time on Race Day.

5. Project Boundaries:

Our deadline is first week of April (02/04/2026) (Week 8).

No budget has been given.

The lecturer should be able to view code through GitHub or the website portfolio.

6. Pre-conditions

- Use only the provided Arduino hardware programmed via Arduino Legacy IDE 1.8.x.
- Write all code in C/C++
- Make sure the BattleBot operates fully autonomously during the relay games.
- Program the bot to avoid all collisions with objects or other robots.

- Documentation containing the MoSCoW analysis, I/O lists, and technical designs.

7. Interim results:

- Project Plan
- Requirements Analysis
- Technical Design
- I/O List
- Portfolio

8. Quality:

The standards, methods and techniques that will be used throughout the project

Action	Methods	Techniques
Design	Design Research	Figma
Code Development	-	C/C++
Electromechanical assembly	-	Arduino Instruction Hand-book
Visualisation	Gantt chart	Excel
Communication	BECD method (giving feedback)	WhatsApp, Teams
Documentation	Work Document, Meeting Minutes	Word, GIT
Norms	Code of Conduct	All team members are foreseen to be available during working hours for the entirety of the pro-ject duration

9. Project Activities (see Figure 1 (Gantt Chart), Figure 2 (Project Breakdown) and Figure 3 (Network Diagram) below):

- **Code Of Conduct**
  - Make first Group meeting
  - Book a meeting
  - Create group in WhatsApp
  - Create group in teams
  - Define rules inside our group
  - Define punishments inside our group
  - Put signatures inside document
  - Send it to Miguel
  - Receive Feedback from Miguel
  - Change document based on feedback

- **Action Plan**
  - Read Module book
  - Read Instruction document
  - Determine team roles
  - Determine project goal, result
  - Take requirements from instruction document
  - Determine project quality
  - Determine which Standards, Methods, Techniques we will use
  - Determine project organization
  - Determine interim results
  - Make meeting with Client
  - Book room for the meeting
  - Make Agenda
  - Make project breakdown
  - Determine which activities we will have
  - Calculate how much hours we need to spend for them
  - Make dependencies
  - Create Network diagram
  - Make critical path
  - Make Gantt chart

- **Gather User Requirements**
  - Search through instruction document
  - Make meeting with the client
  - Create agenda
  - Book a room
  - Create stakeholder analysis
  - Create User stories
  - Create Moscow analysis

- **Requirements Analysis**
  - Create DFD Excel Table detailing requirements
  - Create Requirements Catalogue based on analysis and user stories
  - Send it to Client
  - Receive Client Feedback
  - Improve Requirements Analysis based on feedback given

- **Technical Design**
  - Define Problem and Objectives
  - Check Requirements Analysis
  - Establish System Scope and Assumptions
  - Develop the System Architecture
  - Select Components and Technologies
  - Define Interfaces and I/O
  - Define Testing and Validation Strategy

- **Setup Necessary Programs**
  - Receive advice on programs to use
  - Download programs we need to use
  - Practice with the material given

- **Setup Robot’s Wiring**
  - Receive advice on wiring
  - Read through instruction manual
  - Create a standard through Technical Design

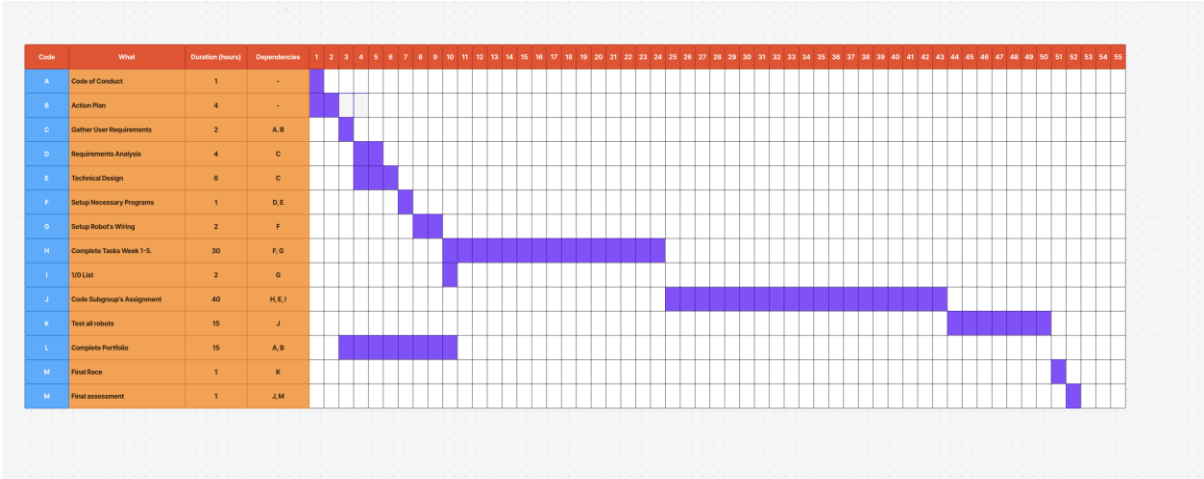
- **Complete Tasks Week 1-5**
  - Studied the basics of C++
  - Studied the C++ code convections
  - Create git hub repository
- **I/O List**
  - Understand System Scope
  - Review Technical Design
  - Identify Input Signals
  - Identify Output Signals
  - Identify Analog I/O
  - Assign Addresses
  - Define Signal Descriptions Clearly
- **Code Subgroups’ Assignments**
  - Choose what subgroup will do which task
  - Write code based on task each group selected
  - Have weekly meetings checking in with the subgroups
- **Test All Robots**
  - Check if each robot successfully finished each route
- **Complete Portfolio**
  - Review other Portfolios for ideas
  - Ask for feedback from lecturers
  - Change Portfolio based on feedback
  - Checked if every aspect of portfolio is written correctly
- **Final Assessment**
  - Make robot complete each individual race
  - Answer the teachers' questions and explain how code works

10. Project Organizations:

Person	Role	Contact Details	Availability	Responsibility
Joel Shoniwa	Team leader, Developer	+31 6 26404215 joel.shoniwa@student.nhlstenden.com	Monday-Friday 8:30 - 17:30	Organizing group meetings, communication with teachers/client, developing line maze part of project.
Ivan Lykhota	Quality Assurance, Developer	+380980577886 ivan.lykhota@student.nhlstenden.com	Monday-Friday 8:30-17:30	Finalizing and ensuring work meets project and client standards, testing products to evaluate quality, developing physical maze part of project
Kyra Kovács	Secretary, Developer	+36 70 391 2200 kyra.kovacs@student.nhlstenden.com	Monday-Friday 8:30-17:30	Writing the meeting minutes, developing line following part of project.
Jehan Master	Developer	+265 993 33 55 04 jehan.master@student.nhlstenden.com	Monday-Friday 8:30-17:30	Developing line following part of project.
Friso Terpstra	Developer	+31 6 11759528	Monday-Friday 8:30-17:30	Developing line maze part of project.

		friso.terp-stra1@stu-dent.nhlstenden.com		
Muhammad Umar Kamran	Developer	+31 6 31377513 muham-mad.umar.kamran@stu-dent.nhlstenden.com	Monday-Friday 8:30-17:30	Developing physical maze part of pro-ject.
Client	Client	+31 6 82803986 mar-tijn.pomp@nhlstenden.com +31 6 51609892 bart.oer-lemans@nhlstenden.com	Monday-Friday 8:30-17:30	Availability at given times for meetings, clarification on pro-ject

11. Planning:



Code	What	Duration (hours)	Dependencies
A	Code of Conduct	1	-
B	Action Plan	4	-
C	Gather User Requirements	2	A, B
D	Requirements Analysis	4	C
E	Technical Design	6	C
F	Setup Necessary Programs	1	D, E
G	Setup Robot's Wiring	2	F
H	Complete Tasks Week 1-5.	30	F, G
I	I/O List	2	G
J	Code Subgroup's Assignment	40	H, E, I
K	Test all robots	15	J
L	Complete Portfolio	15	A, B
M	Final Race	1	K
M	Final assessment	1	J, M

Figure 1 (Gantt Chart)

12. Appendix:

What	Time (hours)	Dependencies	
Code of Conduct	1	-	A
Action Plan	4	-	B
Gather User Requirements	2	A, B	C
Requirements Analysis	4	C	D
Technical Design	6	C	E
Setup Necessary Programs	1	D, E	F
Setup Robot's Wiring	2	F	G
Complete Tasks Week 1-5	30	F, G	H
I/O List	2	G	I
Code Subgroup's Assignment	40	H, E, I	J
Test all robots	15	J	K
Complete Portfolio	15	A,B	L
Final Race	1	K	M
Final assessment	1	J,M	N
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Figure 2 (Project Breakdown)

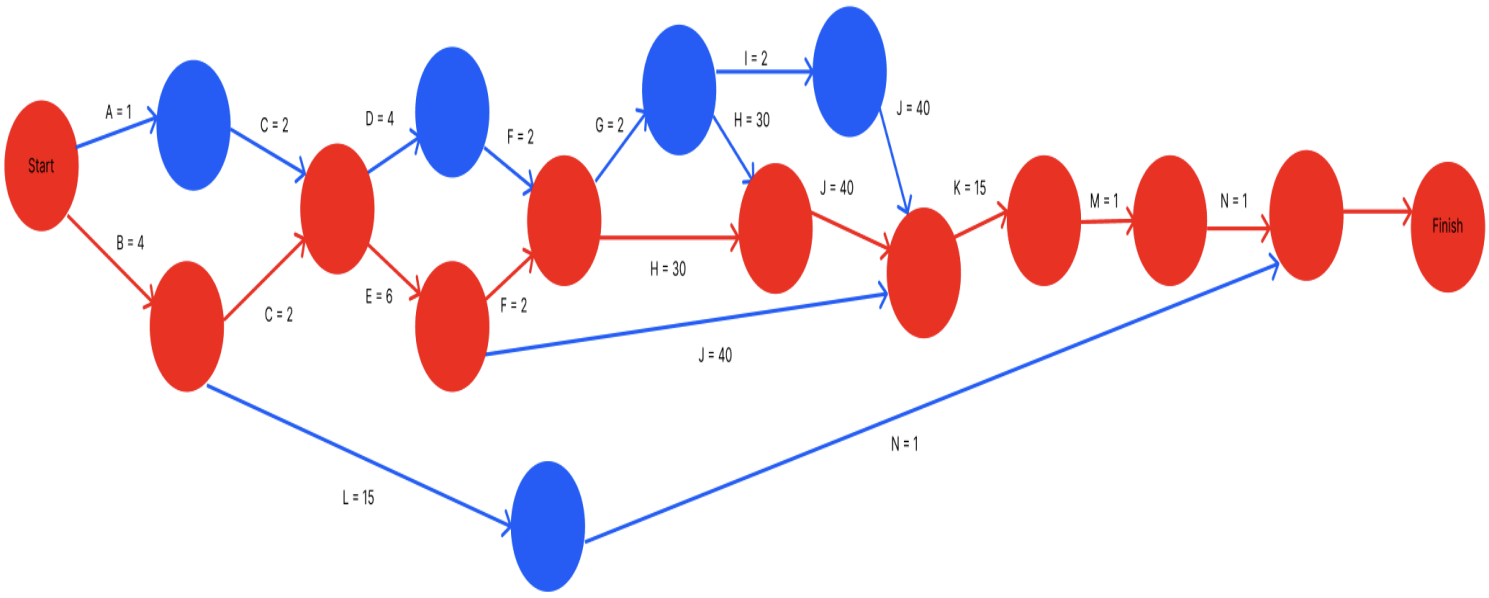

Figure 3 (Network Diagram)